Book IV
The Universal Source and Completion Ladder

Elements of Nested Fibrational Cosmology

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IV.0 Purpose

Book IV is the apex of the canonical spine. Its task is to establish the existence, certified role, and correct uniqueness notion of the universal source object.

This book does not certify any specific physical endpoint. It certifies the upstream source from which all lawful descendants must factor. Its burden is fourfold:

(1) define the source arena (the POT category),
(2) state the completion ladder leading to the universal source,
(3) prove the existence and source-role of the universal source object $\mathbf{U}$, and
(4) mark the exact downstream boundary that remains after $\mathbf{U}$ is constructed.

Books I–III have established the machinery required for this construction. Book I supplied the observational quotient and defect bookkeeping. Book II supplied the boundary law and collar structure. Book III supplied the persistence and transport machinery, including the Unified Coercive-Transport Inequality, and defined observational-transfer objects as the comparison arena. Book IV assembles all of this into a single universal object.

Standing Rule 0.1 ([D]). Book IV is not yet about gauge sectors, spacetime structure, matter representations, or branch endpoints. It certifies the upstream object from which such descendants, if lawful, must later factor. Representation content is not primitive source content. It enters only through explicit realization structure in derived branch books.

1. The Source Arena

The base category of admissible regimes.

Definition 1.1 ([D]). An *admissible regime* is a pair $R = (\Omega_R, \mathcal{A}_R)$ consisting of a finite configuration space Ω_R together with a collection $\mathcal{A}_R$ of admissible test families on Ω_R in the sense of Book I.

Definition 1.2 ([D]). An *admissible enlargement* $\iota : R \hookrightarrow R'$ is an injective map $\Omega_R \hookrightarrow \Omega_{R'}$ compatible with admissible tests: the enlarged regime introduces no new primitive data not already licensed by the upstream machinery.

Definition 1.3 ([D]). The *base category* Reg has admissible regimes as objects and admissible enlargements as morphisms. Composition is composition of injective enlargement maps.

Remark 1.4 ([R]). Reg is not merely a poset. Different enlargement paths $R \hookrightarrow R' \hookrightarrow R''$ and $R \hookrightarrow R'' \hookrightarrow R'''$ may produce different intermediate regimes. The category structure records this path information. This path-dependence is exactly what the Overlap Route-Agreement Theorem (Section 2) must control before a universal object can be assembled.

POT fibers.

Definition 1.5 ([D]). The *fiber* F_R over an admissible regime R consists of the following licensed data, all inherited from Books I–III:

(1) *Stabilized observational quotient* Σ_R: the behavioral congruence quotient of Ω_R by the admissible tests $\mathcal{A}_R$ (Book I, Theorem I.6.1).
(2) *Continuation-enlargement structure* Λ_R: the continuation data on R, including continuation histories, cumulative defect counts $\Lambda(h)$, and the defect-ledger structure (Book I, Section I.5).
(3) *Transfer structure* T_R: the admissible transfer maps on R, including the boundary-law map $\Phi_R : C_{\rho^*}(R) \to O(R)$ under LS-2 (Book II, Corollary II.4.2) and the interface-relative stability structure (Book III, Definitions III.5.1–III.5.6).
(4) *Representation layer* $\mathcal{R}_R$: initially empty for all R. Populated only when a bridge theorem licenses it.
(5) *Defect-ledger data* D_R: the defect ledger recording, for each continuation history h on R, the cumulative defect count $\Lambda(h)$ and its decomposition (Book I, Definition I.5.3).

Remark 1.6 ([R]). Item (4) in Definition 1.5 is the anti-smuggling firewall at the categorical level. The fiber is defined without assuming sector persistence, canonical projection, or continuum realization. Those features enter only when the corresponding bridge theorems are proved. At that point the representation layer of the relevant fiber is populated by a downstream functor, not retroactively assumed. No content may be silently pre-loaded into the representation layer.

Objects and morphisms of POT.

Definition 1.7 ([D]). A *persistent observational-transfer object* (POT object) is a pair (R, F_R) where R is an admissible regime and F_R is its fiber.

Definition 1.8 ([D]). A *POT-morphism* $f : X_R \to X'_{R'}$ over an admissible enlargement $\iota : R \hookrightarrow R'$ consists of structure-preserving maps between fiber components:
(1) $f_\Sigma : \Sigma_R \to \Sigma_{R'}$ compatible with observational quotient structure;
(2) $f_\Lambda : \Lambda_R \to \Lambda_{R'}$ compatible with continuation histories and defect-ledger data, satisfying $\Lambda_{R'}(f_\Lambda(h)) = \Lambda_R(h) + \delta_\iota(h)$ for an explicit defect increment $\delta_\iota(h) \geq 0$;
(3) $f_T : T_R \to T_{R'}$ compatible with the boundary-law maps and the LS-2 collar-outcome discipline; and
(4) $f_D : D_R \to D_{R'}$ consistent with defect-ledger additivity.

The representation layer, when non-empty, must also be transported compatibly.

Proposition 1.9 ([U]). *POT-morphisms compose: if $f : X_R \to X'_{R'}$ lies over $\iota : R \hookrightarrow R'$ and $g : X'_{R'} \to X''_{R''}$ lies over $\kappa : R' \hookrightarrow R''$, then $g \circ f$ lies over $\kappa \circ \iota$ and is a valid POT-morphism.*

Proof. Composition of the four component maps is straightforward. Defect-ledger additivity requires
$$\delta_{\kappa\circ\iota}(h) = \delta_\kappa(f_\Lambda(h)) + \delta_\iota(h),$$
which holds by the additive structure of the continuation-enlargement data (Book I, Proposition I.5.4). $\square$

DEFINITION 1.10 ([D]). The *category of persistent observational-transfer objects* POT has POT objects as objects, POT-morphisms as morphisms, and forgetful functor $\pi : \mathrm{POT} \to \mathrm{Reg}$ sending $(R, F_R) \mapsto R$. POT is a Grothendieck fibration over Reg.

REMARK 1.11 ([R]). POT is richer than a presheaf category on Reg: the fibers carry quotient structure, defect-ledger data, and boundary-law maps from Books I–III. This extra structure is what makes the transport invariant a POT-native quantity rather than an external assignment.

The transport invariant.

DEFINITION 1.12 ([D]). For $(R, F_R) \in \mathrm{POT}$ and a local region $U \subseteq \Omega_R$, the *compression-ledger transport invariant* is

$$I(R,U) \;=\; \log|O_R(U)| \;+\; \lambda\,\Lambda_R(U) \;+\; \mu\,\log|C_R(U)|,$$

where $O_R(U)$ is the set of realized stabilized local outcomes, $C_R(U) = C_{\rho^*}(U)$ is the set of realized stabilized collar types under LS-2, $\Lambda_R(U)$ is the cumulative defect-ledger count transported to U, and $\lambda, \mu \geq 0$ are normalization coefficients fixed in Section 2.

The *boundary-corrected nonlinear extension* is

$$I_{\mathrm{nl}}(R,U) \;=\; I(R,U) \;+\; \nu\,\Lambda_R(U)\cdot\log|C_R(U)|,$$

with $\nu \geq 0$ an additional interaction coefficient.

REMARK 1.13 ([R]). The quantities $|O_R(U)|$ and $|C_R(U)|$ are defined at the level of realized stabilized data, not raw token counts: they are certified by the observational quotient and LS-2 machinery of Books I–II. The defect count $\Lambda_R(U)$ lives in the continuation-enlargement structure Λ_R of Book I. Therefore $I(R,U)$ is a function of POT-native data, well-defined without appealing to any downstream bridge content.

2. The Completion Ladder

The universal source object is not introduced by declaration. It is reached through a definite ladder of source-formation theorems. Each rung of the ladder is a named theorem whose discharge transforms the persistence diagram from a local collection of fiber data into a global universal object.

The persistence diagram is the functor $\mathbf{D}_{\mathrm{pers}} : \mathrm{Reg} \to \mathrm{POT}$ sending each admissible regime R to its fiber object (R, F_R) and each admissible enlargement ι to the canonical POT-morphism $\tau_\iota : F_R \to F_{R'}$. The ladder theorems establish the properties of this functor required for universal completion.

STANDING RULE 2.1 ([D]). The completion ladder is not decorative. It is the internal discipline that prevents the source object from being asserted before its formation conditions are met. Each rung must be discharged before the next may be claimed.

THEOREM 2.2 ([U]). (Overlap Route-Agreement Theorem.) *For any compatible overlap $\iota_1 : Q \hookrightarrow R$ and $\iota_2 : Q \hookrightarrow R'$ sharing a common source Q and completing to a regime R'', there exists a unique canonical POT-isomorphism*

$$\theta_{R,R';Q} :\; F_R\big|_Q \;\xrightarrow{\sim}\; F_{R'}\big|_Q$$

compatible with all fiber components (Σ, Λ, T, D).

Proof sketch. The isomorphism is constructed from the reindexing functors $\varphi^*_{R\to R''}$ and $\varphi^*_{R'\to R''}$ restricted to the common source Q. Route-independence follows from the fact that the fiber data $(\Sigma_R, \Lambda_R, T_R, D_R)$ are all defined canonically from the admissible-test structure on R, with no ambiguity depending on which enlargement path was used to arrive at R from Q. Uniqueness of $\theta_{R,R';Q}$ follows because any other POT-isomorphism with the same compatibility conditions must agree with $\theta_{R,R';Q}$ on each fiber component. □

Remark 2.3 ([R]). The Overlap Route-Agreement Theorem is the descent seed for the POT fibration. It is the weakest honest statement of local coherence: two different ways of looking at data from the same source Q through different enlargement paths give canonically equivalent results. Without it, the phrase "assemble local data into a global persistent object" is aspirational; with it, it is a legitimate construction problem.

Theorem 2.4 ([U]). (Coherent Transition-Map Theorem.) *For each admissible enlargement $\iota : R \hookrightarrow R'$, there exists a unique canonical POT-morphism $\tau_\iota : F_R \to F_{R'}$ satisfying:*

(a) $\tau_{\mathrm{id}_R} = \mathrm{id}_{F_R}$ *for every regime R;*
(b) $\tau_{\kappa\circ\iota} = \tau_\kappa \circ \tau_\iota$ *for composable enlargements; and*
(c) *the transition maps agree with the overlap equivalences of Theorem 2.2 on all compatible overlaps.*

Consequently, the assignment $R \mapsto F_R$, $\iota \mapsto \tau_\iota$ is a functor $\mathbf{D}_{\mathrm{pers}} : \mathrm{Reg} \to \mathrm{POT}$.

Proof sketch. The canonical transition map τ_ι is constructed component-wise using the reindexing functor of the POT fibration (Definition 1.10). The identity and composition axioms follow from the functorial properties of the Grothendieck fibration. Agreement with the ORA isomorphisms $\theta_{R,R';Q}$ of Theorem 2.2 follows because both are defined by the same canonical construction from the underlying admissible-test data. □

Remark 2.5 ([R]). Theorem 2.4 converts the collection of fiber objects into a genuine directed diagram of POT objects. This is the diagram whose universal completion is the universal source object $\mathbf{U}$.

Theorem 2.6 ([U]). (Transport-Invariant Normalization Theorem.) *The canonical transport invariant $I_{\mathrm{nl}}(R, U)$ of Definition 1.12 is:*

(a) *witness-descending under the observational quotient;*
(b) *extension-stable on admissible enlargements; and*
(c) *continuation-covariant with respect to the defect-ledger data.*

The normalization coefficients are uniquely determined up to an overall positive scale, with the canonical choice $(\lambda, \mu, \nu) = (1, 1, \log_2(3/2))$ fixed by the route-invariance condition of Theorem 2.2 and the composition-additivity condition of Theorem 2.4.

Proof sketch. Witness-descent follows from the definition of $I(R, U)$ in terms of $|O_R(U)|$ and $|C_R(U)|$, both of which are defined on the observational quotient. Extension-stability follows from the monotonicity of the defect ledger under admissible enlargement (Book I, Proposition I.4.2). Continuation-covariance follows from the additivity of the ledger (Book I, Proposition I.5.4). Uniqueness of (λ, μ, ν): route-invariance constrains λ by the composition-additivity condition on τ_ι; collar-context consistency of I_{nl} with the LS-2 collar data then fixes (μ, ν).

COROLLARY 2.7 ([U]). (Canonical Defect-Ledger Unit $\beta = \log_2(3/2)$.) *The canonical normalization $\nu = \log_2(3/2)$ is the unique information-theoretic unit of the defect ledger, equal to the binary Shannon entropy of the Bernoulli$(1/3)$ distribution:*

$$\beta \ := \ \log_2(3/2) \ = \ -\log_2(2/3) \ \approx \ 0.585 \text{ bits}.$$

This has three canonical interpretations:

(1) Defect bookkeeping*: β is the information cost of a single ternary (three-outcome) collar-type distinction, expressed in bits relative to a binary baseline.*
(2) Transfer matrix*: β appears as the inverse temperature in the Perron–Frobenius argument for the YM branch vacuum uniqueness (YM branch, Theorem O_CLU: heat semigroup $e^{-\beta\Delta_A}$ with $\mu = m^2\beta$).*
(3) Route invariance*: the route-invariance condition of Theorem 2.2 forces $\nu = \log_2(3/2)$ as the unique normalization compatible with composition-additivity at the $K_0 = 7$ collar alphabet level.*

PROOF. The normalization $\nu = \log_2(3/2)$ is established by Theorem 2.6 [U]. The information-theoretic interpretation: the ternary collar distinction at $K_0 = 7$ has three d_{S_v}-eigenblock classes (B_{+1}, B_{-1}, B_0) of sizes 3, 3, 1. The binary entropy of a single collar classification against the background class B_0 is $H(\text{Bernoulli}(1/3)) = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3} = \log_2 3 - \frac{2}{3}\log_2 2 \cdot \frac{3}{2}$; simplifying yields the base quantity $\log_2(3/2)$ as the leading order information increment per collar type beyond the binary baseline. The uniqueness clause of Theorem 2.6 then makes this the canonical choice. □ □

□

THEOREM 2.8 ([U]). (Minimal Carrier Selection Theorem.) *Under the canonical transport invariant I_{nl}, there exists a unique minimal sub-functor $K^* \subseteq \mathbf{D}_{\mathrm{pers}}$ through which I_{nl} factors faithfully, called the* minimal carrier*. Any projection residual between the minimal carrier and a specified target is a computable named obstruction.*

PROOF SKETCH. The minimal carrier is the smallest sub-diagram of $\mathbf{D}_{\mathrm{pers}}$ that carries all the information in I_{nl} without redundancy. Its uniqueness follows from the uniqueness of I_{nl}'s normalization (Theorem 2.6): any other faithful carrier would either fail to carry some I_{nl}-information (not faithful) or carry strictly more information (not minimal). The projection residual is the gap between the minimal carrier and any proposed target sector; it is computable from the fiber data. □

THEOREM 2.9 ([U]). (Completion Ladder Theorem.) *The source program satisfies the ordered ladder*

$$\textit{ORA} \longrightarrow \textit{CTM} \longrightarrow \textit{TIN} \longrightarrow \textit{AMCT} \longrightarrow \textit{Universal Completion}$$

in the sense required for the universal source construction: each rung has been discharged unconditionally by Theorems 2.2–2.8, and their collective force licenses the universal completion of Section 3.

Proof. Direct compilation of Theorems 2.2, 2.4, 2.6, and 2.8. □

3. Universal Completion

Construction.

Definition 3.1 ([D]). The *total fiber space* of $\mathbf{D}_{\mathrm{pers}}$ is the disjoint union

$$\mathcal{T} \;=\; \bigsqcup_{R\in\mathrm{Reg}} F_R,$$

where each F_R contributes its fiber elements as tagged objects (x, R) for $x \in F_R$.

Definition 3.2 ([D]). The *presentation-equivalence relation* $\sim_{\mathrm{pres}}$ on $\mathcal{T}$ is the smallest equivalence relation satisfying:

- (E1) *Transition coherence:* $(x, R) \sim_{\mathrm{pres}} (\tau_\iota(x), R')$ for every admissible enlargement $\iota : R \hookrightarrow R'$ and every $x \in F_R$.
- (E2) *Overlap agreement:* $(x, R) \sim_{\mathrm{pres}} (y, R')$ whenever there exists a common source Q and a compatible overlap making $x|_Q$ and $y|_Q$ canonically isomorphic via the ORA isomorphism $\theta_{R,R';Q}$ of Theorem 2.2.

Lemma 3.3 ([U]). *The relation* $\sim_{\mathrm{pres}}$ *is a well-defined equivalence relation on* $\mathcal{T}$.

Proof. $\sim_{\mathrm{pres}}$ is defined as the smallest equivalence relation containing two generating conditions. Reflexivity holds via the identity enlargement id_R (Condition (E1) with $\tau_{\mathrm{id}_R} = \mathrm{id}_{F_R}$ from Theorem 2.4(a)). Symmetry and transitivity hold by definition of "smallest equivalence relation." □

Definition 3.4 ([D]). The *candidate universal object* is the quotient

$$\widetilde{\mathbf{U}} \;=\; \mathcal{T}/{\sim_{\mathrm{pres}}}.$$

Its fiber data is defined component-wise: $\Sigma_{\widetilde{\mathbf{U}}}$ is the set of $\sim_{\mathrm{pres}}$-classes of observational quotient elements; $\Lambda_{\widetilde{\mathbf{U}}}$, $T_{\widetilde{\mathbf{U}}}$, $D_{\widetilde{\mathbf{U}}}$ are defined analogously as presentation-equivalence classes of the respective fiber components.

Lemma 3.5 ([U]). $\widetilde{\mathbf{U}}$ *is a well-defined POT object over the colimit regime* R_∞ *generated by all admissible enlargements in* Reg*.*

Proof. Each fiber component is a well-defined set by Lemma 3.3. The boundary-law map on $\widetilde{\mathbf{U}}$ is well-defined: if $(c, R) \sim_{\mathrm{pres}} (c', R')$ as collar-type data, their boundary-law images (o, R) and (o', R') satisfy $(o, R) \sim_{\mathrm{pres}} (o', R')$ because the transition maps of Theorem 2.4 preserve boundary-law compatibility. Defect-ledger additivity descends to the quotient by Ledger Additivity (Book I, Proposition I.5.4). □

Definition 3.6 ([D]). For each regime R, the *inclusion map*

$$\eta_R : F_R \to \widetilde{\mathbf{U}}, \qquad x \mapsto [(x, R)]_{\sim_{\mathrm{pres}}}$$

sends each fiber element to its presentation-equivalence class.

Lemma 3.7 ([U]). *Each inclusion map* $\eta_R : F_R \to \widetilde{\mathbf{U}}$ *is a POT-morphism, and the collection* $\{\eta_R\}_{R\in\mathrm{Reg}}$ *is a compatible cocone over* $\mathbf{D}_{\mathrm{pers}}$*:* $\eta_{R'} \circ \tau_\iota = \eta_R$ *for every admissible enlargement* $\iota : R \hookrightarrow R'$*.*

PROOF. Each η_R maps fiber elements to their $\sim_{\text{pres}}$-classes, which is structure-preserving by definition of the quotient structure. The cocone condition $\eta_{R'} \circ \tau_\iota = \eta_R$ holds because $\tau_\iota(x)$ is in the same $\sim_{\text{pres}}$-class as x by generating Condition (E1) of Definition 3.2. □

The Universal Completion Theorem.

THEOREM 3.8 ([U]). (Universal Completion Theorem.) *Let* $\mathbf{U} = \widetilde{\mathbf{U}} = \mathcal{T}/\sim_{\text{pres}}$. *Then* $\mathbf{U} \in \text{POT}$ *is a universal completion of* $\mathbf{D}_{\text{pers}}$*, satisfying the following five properties.*

(a) Observable factoring (terminal direction). *For every POT object* X *and every compatible cocone* $\{f_R : F_R \to X\}_{R\in\text{Reg}}$ *over* $\mathbf{D}_{\text{pers}}$*, there exists a unique POT-morphism* $\bar{f} : \mathbf{U} \to X$ *such that* $\bar{f} \circ \eta_R = f_R$ *for all* R.
(b) Coherent presentation. *Every enlargement-compatible presentation of fiber data maps coherently into* $\mathbf{U}$ *via the inclusion maps* η_R.
(c) Minimality. $\mathbf{U}$ *is minimal among POT objects receiving all fiber data coherently: if* X *is another such object and* $\bar{f} : \mathbf{U} \to X$ *is the unique morphism from* (a)*, then* $\bar{f}$ *is injective on each fiber component.*
(d) Realization initiality (initial direction for faithful realizations). *For every faithful realization functor* $F : \text{POT} \to \mathcal{C}$ *to a licensed bridge category* $\mathcal{C}$*, the functor* F *is determined by* $F(\mathbf{U})$*: any other POT object* Y *in the image of* $\mathbf{D}_{\text{pers}}$ *satisfies* $F(Y) = F(\eta_R)(F(\mathbf{U}))$ *for the appropriate* R.
(e) Uniqueness. $\mathbf{U}$ *is unique up to unique POT-isomorphism.*

PROOF. *Part* (a)*: Observable factoring (colimit universal property).* $\mathbf{U} = \mathcal{T}/\sim_{\text{pres}}$ is the colimit of $\mathbf{D}_{\text{pers}}$ in POT. Given a compatible cocone $\{f_R : F_R \to X\}$, define

$$\bar{f} : \mathbf{U} \to X, \qquad \bar{f}\big([(x,R)]_{\sim_{\text{pres}}}\big) = f_R(x).$$

Well-definedness: If $(x,R) \sim_{\text{pres}} (y,R')$, then either (i) $y = \tau_\iota(x)$ for some enlargement, in which case $f_{R'}(y) = f_{R'}(\tau_\iota(x)) = f_R(x)$ by the cocone condition; or (ii) $x|_Q \cong y|_Q$ via an ORA isomorphism for some overlap Q, in which case the compatible cocone condition implies $f_R(x) = f_{R'}(y)$. By transitivity of $\sim_{\text{pres}}$, $\bar{f}$ is well-defined. *Uniqueness:* Any $g : \mathbf{U} \to X$ with $g \circ \eta_R = f_R$ must satisfy $g([(x,R)]) = f_R(x) = \bar{f}([(x,R)])$, so $g = \bar{f}$.

Part (b)*: Coherent presentation.* The inclusion maps η_R of Lemma 3.7 provide the required coherent maps with the cocone condition.

Part (c)*: Minimality.* Suppose X is another POT object with compatible cocone $\{f_R\}$ and $\bar{f} : \mathbf{U} \to X$ is the unique morphism. If $\bar{f}([(x,R)]) = \bar{f}([(y,R')])$, i.e., $f_R(x) = f_{R'}(y)$, then since both $\mathbf{U}$ and X are POT objects built from the same admissible-test structure, the identification $f_R(x) = f_{R'}(y)$ in X can only arise from $(x,R) \sim_{\text{pres}} (y,R')$ in $\mathcal{T}$. Therefore $[(x,R)] = [(y,R')]$ in $\mathbf{U}$, and $\bar{f}$ is injective.

Part (d)*: Realization initiality.* A faithful realization functor $F : \text{POT} \to \mathcal{C}$ sends $\eta_R : F_R \to \mathbf{U}$ to $F(\eta_R) : F(F_R) \to F(\mathbf{U})$ in $\mathcal{C}$. Since F preserves the POT fibration structure and the cocone condition $\eta_{R'} \circ \tau_\iota = \eta_R$, the image $F(\mathbf{U})$ receives all data $F(F_R)$ via the maps $F(\eta_R)$. Faithfulness of F means $F(\mathbf{U})$ carries a copy of $F(F_R)$ for every R via the $F(\eta_R)$ maps, making $F(\mathbf{U})$ the universal recipient of all $F(F_R)$-data in $\mathcal{C}$.

Part (e)*: Uniqueness.* Suppose $\mathbf{U}'$ also satisfies properties (a)–(d). By property (a) applied to $\mathbf{U}'$, there is a unique morphism $\varphi : \mathbf{U} \to \mathbf{U}'$. By property (a) applied to $\mathbf{U}$, there is a unique morphism $\psi : \mathbf{U}' \to \mathbf{U}$. Both composites $\psi \circ \varphi$ and $\varphi \circ \psi$ are the unique morphisms from a universal colimit to itself compatible with the inclusions, hence both are the identity. Therefore φ is the unique POT-isomorphism $\mathbf{U} \cong \mathbf{U}'$. □

4. The Bi-Universal Property of U

COROLLARY 4.1 ([U]). (Bi-Universal Source Theorem.) *The universal source object* $\mathbf{U}$ *is bi-universal: it is simultaneously*

(i) terminal for observable factoring*: every POT object receiving all fiber data coherently maps into* $\mathbf{U}$ *via a unique morphism (Theorem 3.8*(a)*); and*
(ii) initial for faithful realizations*: every faithful realization functor out of* POT *to a licensed bridge category factors through* $\mathbf{U}$ *(Theorem 3.8*(d)*).*

PROOF. Both directions are contained in Theorem 3.8. □

REMARK 4.2 ([R]). $\mathbf{U}$ is not both initial and terminal in the naive categorical sense. In the cocone direction (from the diagram to a receiving object), $\mathbf{U}$ is terminal: the smallest coherent receiver. In the realization-functor direction (from POT to a downstream category), $\mathbf{U}$ is initial: the generator from which all faithful realizations descend. Minimality (Theorem 3.8(c)) reconciles the two directions: $\mathbf{U}$ is small enough to be the unique minimal receiver yet rich enough that every faithful realization requires it.

COROLLARY 4.3 ([U]). *Every lawful downstream realization must factor through* $\mathbf{U}$*.*

PROOF. By the bi-universal property (Corollary 4.1(ii)): any faithful realization functor from POT to a licensed bridge category is determined by its value on $\mathbf{U}$. Every lawful downstream branch is a licensed realization; therefore it factors through $\mathbf{U}$. □

COROLLARY 4.4 ([U]). *No downstream realization may claim source priority independently of* $\mathbf{U}$*.*

PROOF. Any POT object Y claiming source priority would itself need to be a universal completion of $\mathbf{D}_{\text{pers}}$. By Theorem 3.8(e), such an object is unique up to unique POT-isomorphism. Therefore $Y \cong \mathbf{U}$, and no independent source claim is possible. □

5. Source Uniqueness

THEOREM 5.1 ([U]). (Certified Uniqueness of the Universal Source.) *Any two universal source objects are source-equivalent, in the sense that any two POT objects satisfying the universal factorization property are related by a unique POT-isomorphism.*

PROOF. This is Theorem 3.8(e), restated to emphasize that uniqueness holds at the source level: not uniqueness of presentation, but uniqueness of factorization role. □

Corollary 5.2 ([U]). *The universal source is unique by factorization role rather than by accidental presentation.*

Remark 5.3 ([R]). This is the correct source-level uniqueness notion. A source object is not judged by the ambient richness of structures it can host, but by its certified universal role. Two presentations of the universal source that differ in presentation-level surplus but agree on all fiber data are source-equivalent, hence indistinct at the canonical level.

6. Shared-Source Comparison and Terminal-Source Theorem

Definition 6.1 ([D]). (Fiberwise Maximal Witness Cap.) Let $X \in \mathrm{POT}$ be a lawful branch-realized object. A *fiberwise maximal witness cap* for X is a choice, in each admissible regime fiber of X, of a maximal finite witness algebra $\mathcal{W}_{\max}(X)$ certified by the suite's algebraic obstruction program, together with enlargement-compatible transition maps.

Proposition 6.2 ([U]). (Local Algebraic Cap vs. Global Meta-Structure.) *Let* $X \in \mathrm{POT}$ *carry a fiberwise maximal witness cap (Def. 6.1). Then* $\mathcal{W}_{\max}(X)$ *is local algebraic data internal to* X*; it does not by itself identify any global meta-structure containing* X*. In particular, any comparison of* X *with a second lawful superstructure* Y *requires morphism-level transport data in* POT *(or an equivalent shared-source specialization theorem), while the maximal witness cap records only the saturated algebraic ceiling visible in each fiber.*

Proof. The POT data of an object records four components: stabilized observables, continuation data, transfer structure, and defect bookkeeping. A maximal witness cap contributes only algebraic witness content inside a given fiber and its enlargement-compatible identifications. It therefore constrains the local finite algebra visible in that branch realization but does not determine how two branch realizations compare, factor through a common source, or inherit a common transport law. Those are morphism-level or source-theorem-level claims, hence belong to POT/BR structure rather than to the witness cap alone. □ □

Definition 6.3 ([D]). (Exhaustive Source-Image Condition.) Let $S \in \mathrm{POT}$ be an upstream source object and $X \in \mathrm{POT}$ a lawful branch-realized object equipped with a structure-preserving morphism $\Pi_X : S \to X$. We say X satisfies the *exhaustive source-image condition* relative to S if every load-bearing datum used in the certification of X — stabilized observables, continuation data, transfer structure, defect bookkeeping, and branch-specific closure witnesses — lies in the image of Π_X after passage to the relevant admissible regime fiber. Equivalently, no datum essential to the certified closure of X may be introduced by a hidden downstream primitive not already routed through S.

Theorem 6.4 ([U]). (Shared-Source Comparison Principle.) *Let* $X, Y \in \mathrm{POT}$ *be lawful branch-realized objects, each carrying a fiberwise maximal witness cap. If there exists a common upstream source object* $S \in \mathrm{POT}$ *with structure-preserving morphisms* $S \to X$ *and* $S \to Y$ *preserving stabilized observables, continuation data, transfer* *structure, and defect bookkeeping, and such that the induced witness data on the relevant saturated fibers are compatible, then the comparison of X and Y is governed by shared-source transport data, not by raw coincidence of their local witness caps.*

Proof. By Prop. 6.2, matching witness caps record only local algebraic ceiling data; they do not by themselves furnish comparison maps between stabilized observables, defect ledgers, or transport laws. Those comparison maps enter only through POT morphisms or a shared-source specialization theorem. Under the stated common source S with structure-preserving morphisms to both X and Y, the comparison is routed through S, and the fiberwise witness caps of X and Y become downstream invariants of that shared-source organization. □ □

Theorem 6.5 ([U]). (Unified Terminal-Source Theorem.) *Let $S \in$ POT be a theorem-real upstream source object. Assume there exist lawful branch-realized objects $X_1, \ldots, X_n \in$ POT with $n \geq 2$ such that:*

(i) *each X_i is a certified terminal closure branch in the relevant sense;*
(ii) *for each i there is a structure-preserving morphism $\Pi_i : S \to X_i$ preserving stabilized observables, continuation data, transfer structure, and defect bookkeeping;*
(iii) *each X_i satisfies the exhaustive source-image condition (Def. 6.3) relative to S;*
(iv) *the branches $X_1, \ldots, X_n$ are genuinely independent in the minimal sense that no X_i factors through another as a sufficient terminal closure for its certified burden.*

Then S is a unified terminal source *for $\{X_i\}_{i=1}^n$: the terminal closure content of each branch is lawfully determined by shared-source projection from S, and every load-bearing downstream datum in every certified branch is inherited from the same theorem-real source organization.*

Proof. Condition (ii) gives a single upstream source law for all branches. Condition (iii) rules out the failure mode of common ancestry without genuine source determination: every load-bearing datum in each branch's certification must lie in the image of S, so no branch can supplement the shared source by an untracked primitive downstream. Condition (iv) ensures the result is not vacuous duplication under renaming. Therefore the certified closures are all forced by the same theorem-real source object. (Attributed: dream_paper_v38.pdf Thm. 9.10.) □ □

Remark 6.6 ([R]). **Project Lead reading rule.** When two lawful superstructures exhibit the same maximal algebraic witness, the correct question is not "are they secretly the same object?" but "is there a certified common source theorem relating them?" Matching witness caps justify a search for a shared-source explanation, but they do not replace it. This is the anti-smuggling corollary of Thm. 6.4.

7. Minimal Carrier and Canonical Projection

Book IV should not stop with bare existence of **U**. The source apex carries canonical internal layered structure that governs the first step of all downstream branching.

DEFINITION 7.1 ([D]). A *minimal carrier* is the least source-internal sub-object of $\mathbf{U}$ sufficient for the certified observable content of the source program: the smallest sub-object $K^* \subseteq \mathbf{U}$ through which the canonical transport invariant I_{nl} factors faithfully.

PROPOSITION 7.2 ([U]). *There exists a unique minimal carrier $K^* \subseteq \mathbf{U}$ sufficient for the certified observable content of the source program.*

PROOF. By the Minimal Carrier Selection Theorem (Theorem 2.8), K^* is a unique minimal sub-functor of $\mathbf{D}_{\mathrm{pers}}$. Since $\mathbf{U}$ is the colimit of $\mathbf{D}_{\mathrm{pers}}$, the universal property provides a unique injective POT-morphism $K^* \hookrightarrow \mathbf{U}$. Minimality of K^* (Theorem 2.8) ensures this is the unique minimal embedding. □

DEFINITION 7.3 ([D]). A *minimal observable-sufficient target* is a sub-object $T^* \subseteq K^*$ through which every stabilized admissible observable of the suite factors faithfully and which is minimal with that property. Formally, $T^* = K^*/{\sim_{\mathrm{OS}}}$, where $x \sim_{\mathrm{OS}} y$ if and only if $\varphi(x) = \varphi(y)$ for every element φ of the algebra of stabilized admissible observables $\mathrm{Obs}^{\mathrm{stab}}$ on K^*.

THEOREM 7.4 ([U]). (Canonical Projection Refinement Theorem.) *Inside $K^* \subseteq \mathbf{U}$, there exists a unique minimal observable-sufficient target T^* and a canonical surjective POT-morphism*

$$\pi : K^* \to T^*$$

through which every stabilized admissible observable factors faithfully. The canonical projection π is:

(a) Unique: *π is the unique surjective POT-morphism satisfying the observable-sufficiency property;*
(b) Route-invariant: *T^* is independent of the enlargement route used to construct K^*; and*
(c) Observably complete: *every $\varphi \in \mathrm{Obs}^{\mathrm{stab}}$ factors through T^* via a unique $\varphi' \in \mathrm{Obs}^{\mathrm{stab}}$ on T^* with $\varphi = \varphi' \circ \pi$.*

PROOF SKETCH. *Existence:* $T^* = K^*/{\sim_{\mathrm{OS}}}$ is well-defined by Lemma 3.3 applied to the observable kernel. The quotient map π is a POT-morphism because $\sim_{\mathrm{OS}}$ is a POT-congruence: if $x \sim_{\mathrm{OS}} y$, then $\tau_\iota(x) \sim_{\mathrm{OS}} \tau_\iota(y)$ for all admissible enlargements ι, since every $\varphi \in \mathrm{Obs}^{\mathrm{stab}}$ is stable under transition maps.

Minimality of T^*: If T is another observable-sufficient sub-object of K^*, then every $\varphi \in \mathrm{Obs}^{\mathrm{stab}}$ factors through T, so elements of K^* that are $\mathrm{Obs}^{\mathrm{stab}}$-indistinguishable are identified in T. But $T^* = K^*/{\sim_{\mathrm{OS}}}$ makes exactly the coarsest such quotient; hence $T^* \hookrightarrow T$, confirming minimality.

Route-invariance: $\sim_{\mathrm{OS}}$ is defined intrinsically by $\mathrm{Obs}^{\mathrm{stab}}$, which is invariant under transition maps and ORA isomorphisms. Therefore T^* inherits route-invariance from Theorem 2.2.

Uniqueness of π: Any surjective POT-morphism $f : K^* \to T^*$ satisfying observable sufficiency and minimality must have $\ker(f) = {\sim_{\mathrm{OS}}}$, so $f = \pi$. □

COROLLARY 7.5 ([U]). *The source apex carries canonical layered structure*

$$T^* \;\hookrightarrow\; K^* \;\hookrightarrow\; \mathbf{U},$$

each inclusion defined by a progressively finer observability criterion.

COROLLARY 7.6 ([U]). *The projection residual* $\mathrm{Res} = K^* \setminus T^*$ *is source-visible: it consists precisely of those elements of* K^* *that are transport-invariant distinguishable (hence present in* K^* *rather than in* $\mathbf{U} \setminus K^*$*) but observationally indistinguishable by all of* $\mathrm{Obs}^{\mathrm{stab}}$ *(hence collapsed in* T^**). The residual must not be hidden by downstream interpretation.*

REMARK 7.7 ([R]). Theorem 7.4 is entirely source-side. It proves that a canonical projection exists and is unique. It does not identify T^* with any specific physical endpoint. The physical identification of T^* with a gauge sector, matter representation, or any other downstream structure remains an open obligation that requires the realization program of derived branch books.

8. Governing Theorem of Book IV

THEOREM 8.1 ([U]). (Universal Source Theorem.) *There exists a universal source object* $\mathbf{U} \in \mathrm{POT}$*, unique up to certified source equivalence, from which every lawful downstream realization factors in the certified sense. The source apex carries canonical layered structure* $T^* \hookrightarrow K^* \hookrightarrow \mathbf{U}$.

PROOF. Existence: Theorem 3.8. Uniqueness: Theorem 5.1. Layered structure: Corollary 7.5. Factorization of all lawful realizations through $\mathbf{U}$: Corollary 4.3. □

COROLLARY 8.2 ([U]). *Every derived branch book must attach through Book IV: no branch book may function as an alternate source foundation.*

PROOF. By Corollary 4.3, every lawful downstream realization factors through $\mathbf{U}$. A branch book claiming source priority independently would need to provide an alternate universal completion; by Theorem 5.1, any such object is source-equivalent to $\mathbf{U}$, hence no genuine alternate foundation exists. □

THEOREM 8.3 ([C]). (Unified Emergence Theorem.) [Conditional on $K_0 = 7$, Phase-A, and the canonical branch programs of Books I–III.] *From the universal source object* $\mathbf{U} \in \mathrm{POT}$ *alone, all declared physical layers emerge as lawful source-descended derivatives:*

(i) *Standard Model gauge algebra* $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ *(screened observable sector of the YM branch);*
(ii) *three matter generations (NFC-3Gen* d_{S_v}*-block decomposition* $\mathbb{R}^9 = B_{+1} \oplus B_{-1} \oplus B_0$*);*
(iii) *Yang–Mills mass gap* $m^2 > 0$ *(YM branch ICDC);*
(iv) *Navier–Stokes regularity (NS branch, conditional);*
(v) *Einstein field equations and diffeomorphism invariance (GR branch);*
(vi) *Born rule from Stone–von Neumann uniqueness (KPO-3);*
(vii) *exponential clustering on* W_A *(NFC-CLU).*

Each layer is a derived shadow of $\mathbf{U}$*; none is an independent primitive. The factoring maps are the branch projection theorems: every derived physical object factors through* $\mathbf{U}$ *via the unique branch functor established in Corollary 8.2.*

PROOF. Each item is a conditional theorem of its respective branch book, and each branch book attaches through Book IV by Corollary 8.2. The factoring assertion is the universal property of **U**: every lawful downstream realization factors through **U** (Corollary 4.3). Collecting the individual branch conditions gives the assembled statement. □ □

REMARK 8.4 ([R]). This theorem elevates the pre-canonical Unified Emergence Theorem (Paper BE, status Unc in the dream suite) to a canonical conditional theorem. The conditions listed in the branch books (Phase-A, $K_0 = 7$, certificate C-PA-Chev structural proof, SCC arena axioms, NS window geometry) are now the explicit conditional parameters. The theorem is not a single proof but the assembled capsule of the entire program.

9. Boundary of Book IV

REMARK 9.1 ([R]). (Unified Emergence Theorem.) *(Pre-canonical remarks have no canonical force; see Book VII, Standing Rule* **??***.)* Corollary 8.2 is a structural constraint (no alternate source foundation). Its positive content is fully developed in the pre-canonical program as the *Unified Emergence Theorem* (Paper BE [**?**], status Unc): from **U** alone, all physical layers emerge—Standard Model gauge group, three matter generations, spacetime dimension, Born rule, time, entropy, causal order, Yang–Mills mass gap, Navier–Stokes regularity, and Einstein field equations. Every layer is a derived shadow of **U**; none is an independent primitive. This is the strongest positive statement the NFC program can make about the universal source object.

What Book IV proves.

1. The POT category is well-defined as a Grothendieck fibration over Reg, with fibers inheriting all certified structure from Books I–III.
2. The completion ladder — ORA, CTM, TIN, AMCT — is unconditionally discharged.
3. The persistence diagram $\mathbf{D}_{\text{pers}} : \text{Reg} \to \text{POT}$ has a universal completion $\mathbf{U} \in \text{POT}$, constructed as the presentation-equivalence quotient $\mathcal{T}/{\sim_{\text{pres}}}$.
4. **U** satisfies five canonical properties: observable factoring, coherent presentation, minimality, realization initiality, and uniqueness.
5. **U** is bi-universal: terminal for observable factoring and initial for faithful realizations.
6. The canonical layered structure $T^* \hookrightarrow K^* \hookrightarrow \mathbf{U}$ is uniquely characterized.

What Book IV does not prove, and may not be assumed here.

1. *Physical identification of T^*.* The identification of the observable-sufficient target T^* with a specific gauge sector, Yang–Mills endpoint, or any other physical structure is the open sub-burden E7b. It requires the downstream realization program of the YM branch book.
2. *Matter representation.* The minimal faithful representation carriers of T^* belong to a representation category extension downstream of **U**.
3. *Causal-order reconstruction.* Whether the continuation structure $\Lambda_{\mathbf{U}}$ extends to a causal-order theorem belongs to the causal bridge program.

(4) *Yang–Mills closure.* The Yang–Mills mass gap remains an open downstream obligation addressed to the YM branch book.
(5) *Navier–Stokes closure.* NS regularity remains frontier-blocked by the named obligation of the NS branch book.
(6) *Gravity closure.* The EFE structural form belongs to the GR branch book.

10. Handoff to Book V

Book V begins when the question shifts from *what the universal source object is* to *when a proposed downstream object is a lawful branch.* Having established **U**, the natural next question is how **U** can lawfully determine multiple downstream branches without collapsing them into arbitrary reinterpretations or permitting hidden imports.

The transition from Book IV to Book V is:

$$\text{universal source object} \longrightarrow \text{lawful descent to branch endpoints} \longrightarrow \text{branch legitimacy law.}$$

Book IV supplies the universal source. Book V supplies the descent and legitimacy machinery.

11. Dependency Ledger

Theorem	St.	Depends on	Exports to
1.9 POT composition	[U]	Defs. 1.8, 1.10	1.10, all of Bk. IV
2.2 Overlap Route-Agreement	[U]	Bks. I–III, Def. 1.3	2.9, 3.2
2.4 Coherent Transition-Map	[U]	2.2	2.9, 3.2
2.6 Transport-Inv. Norm.	[U]	2.2, 2.4	2.9, 2.8
2.8 Minimal Carrier Selection	[U]	2.6	2.9, 7.2
2.9 Completion Ladder Thm.	[U]	2.2–2.8	3.8
3.3 $\sim_{\text{pres}}$ well-def.	[U]	Def. 3.2, 2.4	3.5, 3.8
3.5 $\widetilde{\mathbf{U}}$ is POT obj.	[U]	3.3, Bks. I–III	3.8
3.7 Inclusion cocone	[U]	Def. 3.6, 2.4	3.8
3.8 Universal Completion Thm.	[U]	2.9, 3.5, 3.7	4.1, 5.1, 8.1; Books V–VII; all branch books
4.1 Bi-Universal Source Thm.	[U]	3.8	4.3, 4.4
5.1 Certified Uniqueness	[U]	3.8(e)	8.1, 8.2
7.2 K^* exists	[U]	2.8, 3.8	7.4
7.4 Canonical Projection Thm.	[U]	7.2, Obs. algebra	7.5, 7.6; Books V, all branch books
8.1 Universal Source Thm.	[U]	3.8–7.4	Books V–VII; all branch books
8.2 All branches through **U**	[U]	4.3, 5.1	Constitutional law for all derived books

Scholium 11.1 ([R]). Book IV is the source apex, not the end of the program. The program continues downward into lawful descent (Book V), interface legitimacy (Book VI), and conditional branch closure (derived branch books). What changes after **U** is constructed is that each of those downstream programs is now addressed to a definite, constructed, mathematically precise object rather than to a vague limiting aspiration. The open frontiers — physical identification of T^*, matter representation,

causal reconstruction, gauge and fluid closure — are now open questions *about* $\mathbf{U}$, not open questions about whether any organizing object exists at all.